

Artisan Components, Inc.

TSMC CLN90G (90nm Generic Process) FUSE Release Notes

2004q4v1 (10/10/2004)

- o This is the first release version

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o Process Name
 - 90nm LOGIC SALICIDE (1P9M, 1.0V/2.5V) (CLN90G)
- o Design Rule Document
 - TN90LODR001_1_2.pdf, CMOS Logic Design Rule, v1.2, dated 2004/04/06.
- o Fuse Box Design Rule Document
 - TN90LODR003_0_2.pdf (T-N90-LO-DR-003), v0.2, dated 2004/10/05.
- o GDS Layer Usage Description File
 - TN90LOLE001_02A.pdf, GDS Layer Usage Description File, v0.2A, dated 2003/01/21.
- o DRC Command File (for "d" metal layers = 4, 5, 6, 7, 8, 9)
 - CALIBRE DRC COMMAND FILES -
 - CLN90S_<d>M.12a.1, v1.2a.1, dated 2004/03/15
 - CLN90S_<d>M.01c, Rev0.2a.2, 1-AUG-2002 (recommended rules deck)
 - CFSN90_RDL_<d>M.01c, Rev 0.1c ,17-DEC-2003
 - CFSN90_GENERIC_<d>M.01c, Rev 0.1c ,17-DEC-2003
 - CLN90S_9M.13a.eva060404 VER 1.3a.eva dated 06/04/2004
 - CALIBRE ANTENNA DRC COMMAND FILES --
 - CLN90S_<d>M.ANT.12a.1, Rev v1.2a.1, 2004/03/15
- o LVS Command File
 - TSMC 90nm CALIBRE ERC/LVS COMMAND FILE, V1.1b, dated 2004/03/23.
- o Process Spice Models (used with Synopsys HSPICE -- U-2003.09-SP1)
 - 90nm LOGIC SALICIDE (1P9M, 1.0V/2.5V) (CLN90G), V1.2, dated 2004/02/13.
- o Extractor Technology File
 - T-N90-LO-SP-002-B1, V1.2A, dated 2004/03/04.

EDA Views

This section describes a lists of the EDA tools and versions used to verify this product release. This set of tools and versions corresponds to Artisan EDA Package 3.1.

- o Cadence NC-Verilog
 - 3.4
- o MTI/Mentor ModelSim VHDL
 - 5.7d

- o Synopsys Design Compiler
 - 2002.05
- o Cadence Silicon Ensemble
 - 5.5
- o Adobe Acrobat Reader
 - 4.0
- o Mentor Graphics Calibre
 - 8.8

Technology Implementation

 This section provides information on items that may not be included in the foundry documentation.

Characterization Corners:

The generator is verified on these corners

Fast #1 : FF Process Spice Models, 1.1V, -40 Degrees C.
 Fast #2 : FF Process Spice Models, 1.1V, 0 Degrees C.
 Typical : TT Process Spice Models, 1.0V, 25 Degrees C.
 Slow : SS Process Spice Models, 0.9V, 125 Degrees C.
 Maximum Static Power : FF Process Spice Models, 1.1V, 125 Degrees C.

The executable file name for this fuse Generator is "fuse" and it creates a default instance named "FUSE".

Mandatory Tapeout layers required for foundry:

- GDS2 text layer 63:63 is allocated by TSMC for IP marking.
 *** This is a mandatory tapeout layer ****

Layers not required for mask making:

- GDS2 view generator includes a macro boundary layer (gds2 layer 230:0). This layer is a dummy marker layer used for CAD purposes such as metal density checking or to exclude dummy pattern filling.

Methodology

 This section describes design flow issues that may be applicable to this release.

- o Antenna Modeling
 - Since the fuse-box outputs are connected to diffusion directly, no antenna info is needed.
 - The technology header information is not automatically appended by the generator. If needed, please append this information from the standard cell release.
 - SE currently does not support the new Antenna LEF/DEF syntax. This file should be read into Wroute (2.x.26) standalone before routing and after placement is done.

- o LVS/GDS2

- Fuses are modeled as wires for the LVS purposes.

Known Problems and Solutions

This section describes known problems and work arounds as of the release date of this library.

- Calibre LVS may flag the ERC error "ERC PATHCHK GROUND && ! POWER" for some instances. This error is due to nets which have been purposely connected to ground and not power. These may be ignored.
- Calibre LVS may flag isolated layout nets in layout because of floating guard-ring around fuse.
- Calibre DRC flags following errors
 - a) FUSE.C.1, Min. clearance from fuse window to chip edge 27 um
Needs to be taken care of at chip level.
 - b) CSR.R.2B, Dummy pattern structure must be
Mtop~M1/VIAtop~VIA1/CO/PP/OD.
- Synopsys lib files for Max power corner are not generated. As timing information is not present in Synopsys file you may want to use one of the four files generated for max power corner
